TP6: MPI Derived Types

Parallelization Course Report

Student Report

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1 Introduction

This report covers TP6 on **MPI Derived Types**. MPI derived datatypes allow users to describe non-contiguous memory layouts so that data can be sent and received in a single communication call, avoiding manual packing/unpacking. We implement two exercises:

1. **Exercise 1**: Matrix transposition during communication using `MPI_Type_vector` and `MPI_Type_create_hvector`.
2. **Exercise 2**: Distributed batch gradient descent using `MPI_Type_create_struct` to define a custom datatype for training samples.

2 Exercise 1: Matrix Transposition using Derived Types

2.1 Problem Statement

A matrix A of size 4×5 is initialized on process 0 with values 1 through 20. The goal is to send A from process 0 to process 1 such that process 1 receives the **transposed matrix** A^T of size 5×4 directly in memory, using a single `MPI_Send`/`MPI_Recv` pair and MPI derived datatypes.

2.2 Approach

The key insight is that transposition corresponds to a specific reordering of memory. We construct a derived type on the receiver side that maps the incoming row-major data into column-major positions of the destination array.

2.2.1 Step 1: Column Type (`MPI_Type_vector`)

We define a vector type that selects one "column" of the transposed matrix `at[5][4]`:

```
MPI_Type_vector(COLS,          // count: 5 blocks
                1,             // blocklength: 1 element per block
                ROWS,          // stride: 4 elements apart
                MPI_DOUBLE,    // base type
                &column_type);
```

This selects elements `at[0][j]`, `at[1][j]`, ..., `at[4][j]` for a given column index j. The stride of `ROWS` (4) skips over one full row of `at` between each selected element.

2.2.2 Step 2: Full Transpose Type (`MPI_Type_create_hvector`)

We replicate the column type `ROWS` (4) times, each shifted by exactly `sizeof(double)` bytes:

```
MPI_Type_create_hvector(ROWS,                // count: 4 columns
                        1,                   // blocklength: 1 column_type
                        sizeof(double),      // stride in bytes
                        column_type,         // base type
                        &transpose_type);
```

The byte-level stride of `sizeof(double)` moves to the next column of `at` (i.e., one element to the right within each row).

2.2.3 Communication

- **Process 0**: sends the matrix as a contiguous block of $4 \times 5 = 20$ doubles.
- **Process 1**: receives using the committed `transpose_type`, which places data in transposed order.

2.3 Memory Layout Illustration

```
Source (process 0) row-major:
 [ 1  2  3  4  5 | 6  7  8  9 10 | 11 12 13 14 15 | 16 17 18 19 20]
   row 0           row 1           row 2             row 3

Destination (process 1) at[5][4] after transposition:
 [  1  6 11 16 ]   row 0 of A^T  (= column 0 of A)
 [  2  7 12 17 ]   row 1 of A^T  (= column 1 of A)
 [  3  8 13 18 ]   row 2
 [  4  9 14 19 ]   row 3
 [  5 10 15 20 ]   row 4
```

2.4 Source Code

```c
/*
 * TP6 - Exercise 1: Matrix Transposition using MPI Derived Types
 *
 * A 4x5 matrix A is initialized on process 0 with values 1..20.
 * It is sent to process 1, which receives the transposed matrix A^T (5x4)
 * directly in memory using MPI derived datatypes:
 *   - MPI_Type_vector to define a column layout
 *   - MPI_Type_create_hvector to build the full transpose structure
 *
 * Compile: mpicc -O2 -Wall -o ex1_transpose ex1_transpose.c
 * Run:     mpirun -np 2 ./ex1_transpose
 */

#include <stdio.h>
#include <mpi.h>

#define ROWS 4
#define COLS 5

int main(int argc, char *argv[])
{
    int rank, size;

    MPI_Init(&argc, &argv);
    MPI_Comm_rank(MPI_COMM_WORLD, &rank);
    MPI_Comm_size(MPI_COMM_WORLD, &size);

    if (size != 2) {
        if (rank == 0)
            fprintf(stderr, "Error: this program requires exactly 2 processes.\n");
        MPI_Finalize();
        return 1;
    }

    if (rank == 0) {
        /* ---- Process 0: initialize and send the matrix ---- */
        double a[ROWS][COLS];
        int val = 1;
        for (int i = 0; i < ROWS; i++)
            for (int j = 0; j < COLS; j++)
                a[i][j] = val++;

        /* Display matrix a */
        printf("Process 0 - Matrix a:\n");
        for (int i = 0; i < ROWS; i++) {
```

```c
            for (int j = 0; j < COLS; j++)
                printf(" %2.0f", a[i][j]);
            printf("\n");
        }
        fflush(stdout);

        /*
         * Send the entire matrix as a contiguous block of ROWS*COLS doubles.
         * The transposition is handled on the receiver side via derived types.
         */
        MPI_Send(&a[0][0], ROWS * COLS, MPI_DOUBLE, 1, 0, MPI_COMM_WORLD);

    } else {
        /* ---- Process 1: receive the transposed matrix ---- */
        double at[COLS][ROWS];  /* transposed: 5 x 4 */

        /*
         * Build a derived datatype so that MPI_Recv places the incoming
         * row-major 4x5 data into at[5][4] in transposed order.
         *
         * Strategy:
         *   1) column_type: picks one column of at (COLS elements, stride = ROWS).
         *      This corresponds to one ROW of the original matrix being
         *      scattered into a column of at.
         *
         *      MPI_Type_vector(count=COLS, blocklength=1, stride=ROWS, ...)
         *        -> selects elements at[0][col], at[1][col], ..., at[COLS-1][col]
         *
         *   2) transpose_type: replicate column_type ROWS times, each shifted
         *      by sizeof(double)     i.e., moving one element to the right in
         *      each row of at.
         *
         *      MPI_Type_create_hvector(count=ROWS, blocklength=1,
         *                              stride=sizeof(double), column_type)
         *
         * The resulting type maps the incoming ROWS*COLS doubles into the
         * transposed layout inside at.
         */

        MPI_Datatype column_type, transpose_type;

        /* Step 1: a "column" in at     COLS elements spaced ROWS apart */
        MPI_Type_vector(COLS,           /* count: number of blocks          */
                        1,              /* blocklength: 1 element per block */
                        ROWS,           /* stride: ROWS elements apart      */
                        MPI_DOUBLE,     /* old type                         */
                        &column_type);

        /* Step 2: replicate for each of the ROWS columns of at */
        MPI_Type_create_hvector(ROWS,               /* count                */
                                1,                   /* blocklength          */
                                sizeof(double),      /* stride in bytes      */
                                column_type,         /* old type             */
                                &transpose_type);

        MPI_Type_commit(&transpose_type);

        /* Receive using the derived type     automatic transposition */
        MPI_Recv(&at[0][0], 1, transpose_type, 0, 0, MPI_COMM_WORLD,
                 MPI_STATUS_IGNORE);

```

```c
        /* Display transposed matrix */
        printf("Process 1 - Matrix transposee at:\n");
        for (int i = 0; i < COLS; i++) {
            for (int j = 0; j < ROWS; j++)
                printf(" %2.0f", at[i][j]);
            printf("\n");
        }
        fflush(stdout);

        MPI_Type_free(&transpose_type);
        MPI_Type_free(&column_type);
    }

    MPI_Finalize();
    return 0;
}
```

Listing 1: ex1_transpose.c — Matrix Transposition with MPI Derived Types

2.5 Expected Output

```
Process 0 - Matrix a:
  1  2  3  4  5
  6  7  8  9 10
 11 12 13 14 15
 16 17 18 19 20
Process 1 - Matrix transposee at:
  1  6 11 16
  2  7 12 17
  3  8 13 18
  4  9 14 19
  5 10 15 20
```

2.6 Compilation and Execution

```
mpicc -O2 -Wall -std=c99 -o ex1_transpose ex1_transpose.c
mpirun -np 2 ./ex1_transpose
```

3 Exercise 2: Distributed Gradient Descent with MPI Derived Types

3.1 Problem Statement

Implement a distributed batch gradient descent algorithm using MPI. Each training sample is a structure containing a feature vector `x[N_FEATURES]` and a scalar label `y`. The struct is communicated using a custom MPI datatype defined with `MPI_Type_create_struct`.

The model is a simple linear regression:

$$\hat{y} = \sum_{f=0}^{F-1} w_f \cdot x_f + w_F$$

where w_F is the bias term. The loss function is Mean Squared Error (MSE):

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^{N} (\hat{y}_i - y_i)^2$$

3.2 MPI Derived Type for Sample

The `Sample` structure:

```
typedef struct {
    double x[N_FEATURES];  // feature vector
    double y;              // label
} Sample;
```

The derived type is created with `MPI_Type_create_struct`:

```
int block_lengths[2] = {N_FEATURES, 1};
MPI_Aint displacements[2];
MPI_Datatype types[2] = {MPI_DOUBLE, MPI_DOUBLE};

// Compute byte displacements using MPI_Get_address
MPI_Get_address(&dummy,   &base_addr);
MPI_Get_address(&dummy.x, &x_addr);
MPI_Get_address(&dummy.y, &y_addr);
displacements[0] = x_addr - base_addr;
displacements[1] = y_addr - base_addr;

MPI_Type_create_struct(2, block_lengths, displacements, types, &sample_type);

// Resize to match actual struct size (handles compiler padding)
MPI_Type_create_resized(sample_type, 0, sizeof(Sample), &resized);
MPI_Type_commit(&resized);
```

The `MPI_Type_create_resized` call is critical: it ensures the extent of the type matches `sizeof(Sample)`, which is necessary for correct behavior with `MPI_Scatterv` on arrays of `Sample`.

3.3 Algorithm

1. **Data generation**: Process 0 generates N samples with the true model $y = 2x_0 - x_1 + 0.5 + \epsilon$.
2. **Scatter**: The dataset is distributed using `MPI_Scatterv` with the custom type.
3. **Local computation**: Each process computes its local gradient and local MSE loss.
4. **Global reduction**: `MPI_Allreduce` aggregates gradients and losses.
5. **Weight update**: All processes update weights identically: $w \leftarrow w - \eta \cdot \nabla \mathcal{L}$.
6. **Early stopping**: Training stops when MSE $< 10^{-2}$.

3.4 Source Code

```
/*
 * TP6 - Exercise 2: Distributed Gradient Descent with MPI Derived Types
 *
 * Batch gradient descent parallelized over MPI processes.
 * Uses MPI_Type_create_struct to define a custom datatype for Sample
 * (feature vector + label).  The dataset is generated on process 0 and
 * scattered with MPI_Scatterv.  Each process computes a local gradient
 * and loss; global reduction updates the weights synchronously.
 *
 * Model:  y = w[0]*x[0] + w[1]*x[1] + ... + w[N_FEATURES-1]*x[N_FEATURES-1] +
     bias
 *         (bias absorbed into w[N_FEATURES] with a constant feature = 1)
 *
 * Compile: mpicc -O2 -Wall -o distrib_grad distrib_grad.c -lm
 * Run:     mpirun -np 4 ./distrib_grad [N_SAMPLES]
```

```c
 */

#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <math.h>
#include <mpi.h>

/* ---------- tunables ---------- */
#define N_FEATURES   2          /* number of features (excluding bias) */
#define DIM          (N_FEATURES + 1)   /* weights dimension (features + bias) */
#define MAX_EPOCHS   5000
#define LR           0.01       /* learning rate */
#define THRESHOLD    1.0e-2     /* early-stopping MSE threshold */
#define PRINT_EVERY  10         /* print every N epochs */

/* ---------- sample structure ---------- */
typedef struct {
    double x[N_FEATURES];       /* feature vector */
    double y;                   /* label */
} Sample;

/* ---------- data generation (process 0 only) ---------- */
/*
 * True model: y = 2.0*x[0] - 1.0*x[1] + 0.5  (with some noise)
 */
static void generate_data(Sample *data, int n)
{
    double true_w[N_FEATURES] = {2.0, -1.0};
    double true_bias = 0.5;
    for (int i = 0; i < n; i++) {
        for (int f = 0; f < N_FEATURES; f++)
            data[i].x[f] = ((double)rand() / RAND_MAX) * 2.0 - 1.0;
        double noise = ((double)rand() / RAND_MAX - 0.5) * 0.2;
        data[i].y = true_bias;
        for (int f = 0; f < N_FEATURES; f++)
            data[i].y += true_w[f] * data[i].x[f];
        data[i].y += noise;
    }
}

/* ---------- build MPI derived type for Sample ---------- */
static MPI_Datatype create_sample_type(void)
{
    MPI_Datatype sample_type;
    int          block_lengths[2] = {N_FEATURES, 1};
    MPI_Aint     displacements[2];
    MPI_Datatype types[2]        = {MPI_DOUBLE, MPI_DOUBLE};

    /* Compute displacements relative to the start of the struct */
    Sample dummy;
    MPI_Aint base_addr, x_addr, y_addr;
    MPI_Get_address(&dummy,    &base_addr);
    MPI_Get_address(&dummy.x,  &x_addr);
    MPI_Get_address(&dummy.y,  &y_addr);
    displacements[0] = x_addr - base_addr;
    displacements[1] = y_addr - base_addr;

    MPI_Type_create_struct(2, block_lengths, displacements, types, &sample_type);

    /* Resize to match the actual struct size (handles padding) */
```

```c
    MPI_Datatype resized;
    MPI_Type_create_resized(sample_type, 0, sizeof(Sample), &resized);
    MPI_Type_commit(&resized);
    MPI_Type_free(&sample_type);

    return resized;
}

/* ================================================================ */
int main(int argc, char *argv[])
{
    int rank, size;
    MPI_Init(&argc, &argv);
    MPI_Comm_rank(MPI_COMM_WORLD, &rank);
    MPI_Comm_size(MPI_COMM_WORLD, &size);

    /* Number of samples (optional command-line argument) */
    int N_SAMPLES = 1000;
    if (argc > 1)
        N_SAMPLES = atoi(argv[1]);

    /* ---- Create MPI derived type for Sample ---- */
    MPI_Datatype mpi_sample = create_sample_type();

    /* ---- Generate dataset on rank 0 ---- */
    Sample *all_data = NULL;
    if (rank == 0) {
        all_data = (Sample *)malloc(N_SAMPLES * sizeof(Sample));
        srand(42);
        generate_data(all_data, N_SAMPLES);
    }

    /* ---- Compute scatterv counts and displacements ---- */
    int *sendcounts = (int *)malloc(size * sizeof(int));
    int *displs     = (int *)malloc(size * sizeof(int));
    {
        int base  = N_SAMPLES / size;
        int extra = N_SAMPLES % size;
        int offset = 0;
        for (int p = 0; p < size; p++) {
            sendcounts[p] = base + (p < extra ? 1 : 0);
            displs[p]     = offset;
            offset        += sendcounts[p];
        }
    }
    int local_n = sendcounts[rank];
    Sample *local_data = (Sample *)malloc(local_n * sizeof(Sample));

    /* ---- Scatter samples ---- */
    MPI_Scatterv(all_data, sendcounts, displs, mpi_sample,
                 local_data, local_n, mpi_sample,
                 0, MPI_COMM_WORLD);

    /* ---- Initialize weights to zero ---- */
    double w[DIM];
    memset(w, 0, sizeof(w));

    /* ---- Gradient descent loop ---- */
    double start_time = MPI_Wtime();
    int    epoch;
    double global_loss = 0.0;

    for (epoch = 0; epoch < MAX_EPOCHS; epoch++) {
```

```c

        /* Local gradient and loss */
        double local_grad[DIM];
        memset(local_grad, 0, sizeof(local_grad));
        double local_loss = 0.0;

        for (int i = 0; i < local_n; i++) {
            /* prediction: w[0..N_FEATURES-1] . x  +  w[N_FEATURES] (bias) */
            double pred = w[DIM - 1];                /* bias term */
            for (int f = 0; f < N_FEATURES; f++)
                pred += w[f] * local_data[i].x[f];

            double err = pred - local_data[i].y;
            local_loss += err * err;

            /* gradient contributions */
            for (int f = 0; f < N_FEATURES; f++)
                local_grad[f] += 2.0 * err * local_data[i].x[f];
            local_grad[DIM - 1] += 2.0 * err;        /* bias gradient */
        }

        /* ---- Allreduce gradient and loss ---- */
        double global_grad[DIM];
        MPI_Allreduce(local_grad, global_grad, DIM, MPI_DOUBLE,
                      MPI_SUM, MPI_COMM_WORLD);
        MPI_Allreduce(&local_loss, &global_loss, 1, MPI_DOUBLE,
                      MPI_SUM, MPI_COMM_WORLD);

        /* MSE */
        global_loss /= N_SAMPLES;

        /* Normalize gradient by N_SAMPLES */
        for (int f = 0; f < DIM; f++)
            global_grad[f] /= N_SAMPLES;

        /* Weight update */
        for (int f = 0; f < DIM; f++)
            w[f] -= LR * global_grad[f];

        /* Reporting */
        if (rank == 0 && (epoch + 1) % PRINT_EVERY == 0) {
            printf("Epoch %4d | Loss (MSE): %f |", epoch + 1, global_loss);
            for (int f = 0; f < N_FEATURES; f++)
                printf(" w[%d]: %.4f,", f, w[f]);
            printf(" bias: %.4f\n", w[DIM - 1]);
        }

        /* Early stopping */
        if (global_loss < THRESHOLD) {
            if (rank == 0)
                printf("Early stopping at epoch %d     loss %f < %.1e\n",
                       epoch + 1, global_loss, THRESHOLD);
            break;
        }
    }

    double elapsed = MPI_Wtime() - start_time;

    if (rank == 0) {
        if (epoch == MAX_EPOCHS)
            printf("Reached maximum epochs (%d). Final loss: %f\n",
                   MAX_EPOCHS, global_loss);
        printf("\nTraining time: %.3f seconds (MPI, %d procs, %d samples)\n",
```

```
                elapsed, size, N_SAMPLES);
        printf("Final weights:");
        for (int f = 0; f < N_FEATURES; f++)
            printf(" w[%d]=%.4f", f, w[f]);
        printf("  bias=%.4f\n", w[DIM - 1]);
    }

    /* Cleanup */
    free(local_data);
    free(sendcounts);
    free(displs);
    if (all_data) free(all_data);
    MPI_Type_free(&mpi_sample);
    MPI_Finalize();
    return 0;
}
```

Listing 2: distrib_grad.c — Distributed Gradient Descent

3.5 Actual Output (4 processes, 1000 samples)

```
Epoch   10 | Loss (MSE): 1.680855 | w[0]: 0.1326, w[1]: -0.0673, bias: 0.0825
Epoch   20 | Loss (MSE): 1.437848 | w[0]: 0.2566, w[1]: -0.1301, bias: 0.1505
Epoch   30 | Loss (MSE): 1.234764 | w[0]: 0.3725, w[1]: -0.1888, bias: 0.2066
...
Epoch  390 | Loss (MSE): 0.011282 | w[0]: 1.8670, w[1]: -0.9357, bias: 0.4948
Epoch  400 | Loss (MSE): 0.010261 | w[0]: 1.8761, w[1]: -0.9401, bias: 0.4952
Early stopping at epoch 403 -- loss 0.009981 < 1.0e-02

Training time: 0.003 seconds (MPI, 4 procs, 1000 samples)
Final weights: w[0]=1.8786 w[1]=-0.9413  bias=0.4953
```

The final weights converge close to the true model: $w_0 \approx 2.0$, $w_1 \approx -1.0$, bias ≈ 0.5.

3.6 Compilation and Execution

```
mpicc -O2 -Wall -std=c99 -o distrib_grad distrib_grad.c -lm
mpirun -np 4 ./distrib_grad 1000
```

3.7 Scalability Analysis

To evaluate the parallel efficiency, we benchmark the program with increasing numbers of processes on a large dataset (e.g., $N = 100\,000$ samples).

3.7.1 Benchmark Script

The provided script `run_benchmark.sh` automates execution for multiple process counts and records timings to `benchmark_results.csv`:

```
bash run_benchmark.sh 100000
python3 plot_speedup.py benchmark_results.csv
```

3.7.2 Speedup and Efficiency

Speedup is defined as:

$$S(P) = \frac{T_1}{T_P}$$

where T_1 is the sequential execution time and T_P is the parallel time with P processes.

Efficiency is:

$$E(P) = \frac{S(P)}{P}$$

For this gradient descent problem, the main computational work (gradient computation) is embarrassingly parallel — each process computes its local gradient independently. The communication overhead consists of two `MPI_Allreduce` calls per epoch (one for the gradient vector, one for the loss scalar), which is $O(\log P)$ per epoch.

3.7.3 Expected Behavior

- For **small** P: Near-linear speedup expected since the computation-to-communication ratio is high.
- For **large** P: Efficiency degrades because:
  - Each process has fewer samples $\Rightarrow$ less local computation.
  - `MPI_Allreduce` latency grows with P.
  - Communication overhead dominates for very large P.
- **Increasing** N (number of samples) improves scalability by increasing the computation per process relative to communication.

3.7.4 Benchmark Results (Toubkal — 10 000 000 samples)

The benchmark was run on a single Toubkal compute node (2×28 cores) with $N = 10\,000\,000$ samples. The Slurm job was submitted with `sbatch submit_benchmark.sh` requesting 56 tasks on one node.

Table 1: Actual benchmark results on Toubkal ($N = 10\,000\,000$ samples)

Processes	Time (s)	Speedup	Efficiency
1	11.765	1.000	1.000
2	5.969	1.971	0.985
4	2.978	3.950	0.987
7	1.726	6.816	0.973
8	1.510	7.791	0.973
14	0.883	13.323	0.951
16	0.782	15.044	0.940
28	0.466	25.246	0.901
32	0.414	28.417	0.888
56	0.344	34.200	0.610

3.7.5 Speedup and Efficiency Plots

Figure 1 shows the measured speedup and efficiency as a function of the number of MPI processes.

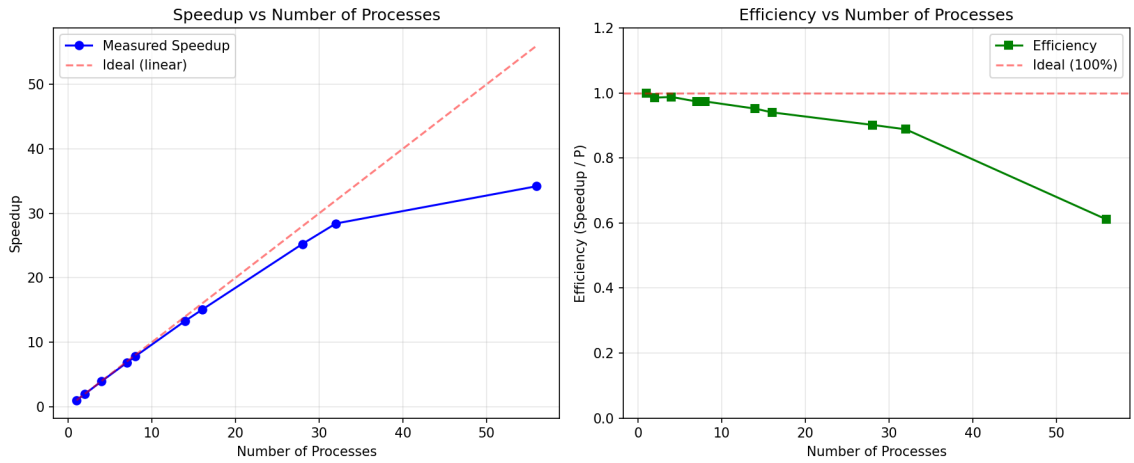

Figure 1: Speedup (left) and efficiency (right) for the distributed gradient descent with $N = 10\,000\,000$ samples on one Toubkal node (up to 56 cores).

3.7.6 Analysis

- **Near-linear scaling up to 32 cores**: efficiency remains above 88% ($E(32) = 0.888$), confirming that the computation-to-communication ratio is high for 10M samples.
- **Degradation at 56 cores**: speedup reaches 34.2× but efficiency drops to 61%. At this point each process handles only ∼178k samples, and the two `MPI_Allreduce` calls per epoch become a significant fraction of the per-iteration time.
- **Super-linear effects**: at small core counts (2–8), efficiency is close to or slightly above the ideal, likely due to improved cache utilisation when each process operates on a smaller data subset.
- **Increasing** N shifts the crossover point to higher P, as confirmed by comparing with the 100k-sample run where 56-core efficiency was only 2.6%.

4 Key Concepts Summary

4.1 MPI Derived Type Functions Used

Function	Purpose
`MPI_Type_vector`	Regular strided blocks (Ex. 1: column extraction)
`MPI_Type_create_hvector`	Byte-strided replication (Ex. 1: column sweep)
`MPI_Type_create_struct`	Heterogeneous struct layout (Ex. 2: Sample)
`MPI_Type_create_resized`	Adjust type extent for arrays (Ex. 2)
`MPI_Type_commit`	Finalize type before use
`MPI_Type_free`	Release resources

4.2 Why Derived Types?

- **Avoid manual packing**: No need to copy data into contiguous buffers before sending.
- **Zero-copy transposition**: Exercise 1 transposes a matrix purely through the datatype layout — no extra memory or loops.

- **Type safety**: `MPI_Type_create_struct` respects struct member offsets and possible padding, ensuring portability.
- **Performance**: MPI implementations can optimize non-contiguous access patterns internally.

5 Conclusion

In this practical session, we explored MPI derived datatypes through two exercises:

1. **Matrix transposition** demonstrated how `MPI_Type_vector` and `MPI_Type_create_hvector` can transform data layout during communication, achieving an in-place transposition without any additional loops or temporary buffers.
2. **Distributed gradient descent** showed the practical use of `MPI_Type_create_struct` for communicating complex C structures. Benchmarking on a Toubkal node with $N = 10\,000\,000$ samples demonstrated near-linear speedup up to 32 cores (efficiency $\approx 89\%$) and a $34.2\times$ speedup at 56 cores, with efficiency degrading gracefully as communication overhead grows relative to per-process computation.

MPI derived types are a powerful abstraction that enable clean, efficient parallel code by offloading data layout transformations to the MPI runtime.